Assessment Brief

Module title:	Programming for Data Analytics
Module code:	COM 7024
Assignment title:	Project
Assignment format:	MS Word / PDF
Word/time limit:	3000 words
File type	. docx / .pdf
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Rubric/Marking Matrix	A set of rules or guidelines used to grade or assess work.

Task summary:

As part of the formal assessment for the programme you are required to submit a **Programming for Data Analytics** assignment. Please refer to your Student Handbook for full details of the programme assessment scheme and general information on preparing and submitting assignments. The assignment brief will specifically give details and instructions for the assignment. No examination, or details of, are included within this module.

Assignment instructions:

Description: The assignment is given as one extended task, unless multiple components have been given. Task 1 has been designed to check your understanding of Data Analytics programming methodologies for critical analysis, through a real-world mini–Data Analytics project. All of which will be written in Python. Plus, Task 1 allows you to demonstrate your reporting skills, comprehensively written in an easily understandable manner, based on your Data Analytics study.

A clear, concise analysis for the Task is to be given within the submission, complimented with screenshot evidence of all processes and results. You are to submit a single Word document for the task, and your code for Task 1 is to be included within an appendix so that it can be checked and verified it be working correctly. Your student ID number must be clearly defined in the uploaded file.

Assignment Task

Task 1 – Data Analytics in Python

Detail Task 1

You have recently acquired an analyst position at One Time Sales and Marketing Ltd. One of the sales managers has asked you to perform exploratory data analysis, using Python, on a sales tracking data set: the focus being on the *Sales Person* and *Total sales value* over the given time period of the data set, plus a further study on whether there is a correlation between *priority* and *value* over the same period. The purpose of the study is to appraise analytical methodologies to help predict possible sales trends in the future for the identified sales representatives.

Therefore, you are to correctly import the sales data set named ***Sales Data PDA 4052.csv*** and perform initial statistical tests on all appropriate variables, once ensuring all data types are correct. Initial process now completed, perform all appropriate pre-processing methodologies, which you deem are necessary to enable a further statistical study, demonstrating improvements from the pre-processing stage.

Based upon the focused variables above, critically evaluate and conduct what you determine are appropriate further investigations, with graphical representation, so that a meaningful study can be presented to your sales manager. You are to give full written reasoning and justification to complement each procedure carried out throughout the entire investigation.

(Total 100 marks)

(LO's:1,2, 3 & 4)

(2000 Words Equivalence – Coded Work)

(1000 Words Equivalence – Reflective report)

(Note: In this assignment, we will not select or apply any ML model, only apply preprocessing techniques including EDA, and write a reflective report based on the findings and at the end suggest some insightful recommendations that would help for further analysis and support in the decision-making process.)

End of Questions

As technology and platforms may change, your module tutor will provide you with up-to-date details.

Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

1. Demonstrate a critical understanding of the concepts and principles of programming and programming paradigms. [**LO 1]**
2. Employ relevant tools, programmatic techniques, and features to prepare, manipulate, and analyse data. [**LO 2]**
3. Critically evaluate the preliminary results of manipulating and analysing data **[LO 3]**
4. Evaluate and apply digital tools and/or services while critically reflecting on opportunities for developing novel digital capabilities. Identify, select, plan for, use, modify, and evaluate digital applications and strategies to enhance the achievement of aims and desired outcomes. [**LO 4]**

Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

Guidelines and policies

You can find links to more useful information about the assignment and university policies below.

Word/time limit policy	Click here to view the Arden University word count/time limit policy
Referencing guidelines	
Please follow the referencing guidelines that are appropriate for your degree programme. If you are unsure which you should be using, please contact your module team.	Click here for Harvard referencing guidelines
Academic integrity and misconduct policy	Click here to view Arden University's policy on academic integrity and misconduct
Statement on use of artificial intelligence on assessment	Click here to view Arden University's statement on the use of artificial intelligence on assessment
Support information	Click here to view guidance on how to apply for short-term extensions
	Click here to view guidance on how to apply for extenuating circumstances